

Package ‘rWind’

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Type Package

Title Download, Edit and Include Wind and Sea Currents Data in
Ecological and Evolutionary Analysis

Version 1.2.0

BugReports <https://github.com/jabiologo/rWind/issues>

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Description Tools to download and manage surface wind and sea-current data
from the Global Forecast System
<<https://www.ncei.noaa.gov/products/weather-climate-models/global-forecast>>
and to compute flow-sensitive connectivity between locations.

URL <https://github.com/jabiologo/rWind>,
<https://allthiswasfield.blogspot.com/>

License GPL (>=3)

LazyData true

Imports raster (>= 2.5-8), terra, gdistance, Matrix, lubridate

Suggests testthat, rmarkdown, knitr

VignetteBuilder knitr

Depends R (>= 3.5)

RoxygenNote 7.3.3

Roxygen list(markdown = TRUE)

NeedsCompilation no

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rWind-package	<i>Download, edit and include wind and sea currents data in ecological and evolutionary analysis</i>
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Description

rWind contains tools for downloading, editing and transforming wind and sea currents data from the Global Forecast System (GFS) and Ocean Surface Current Analyses Real-time (OSCAR). It also allows wind and sea currents data to be used to compute minimum-cost paths from flow speed and direction and to perform connectivity analysis.

Details

The complete list of functions can be displayed with `library(help = rWind)`. For more information, please check: <https://allthiswasfield.blogspot.com/>.

Author(s)

Javier Fernández-López
Klaus Schliep
Yurena Arjona
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See Also

Useful links:

- <https://github.com/jabiologo/rWind>
- <https://allthiswasfield.blogspot.com/>
- Report bugs at <https://github.com/jabiologo/rWind/issues>

arrowDir

Arrow direction fitting for Arrowhead function from "shape" package

Description

arrowDir adapts wind direction value to be used by Arrowhead function from "shape" package to plot wind direction for each coordinate.

Usage

```
arrowDir(W)
```

Arguments

W	An object of class rWind or a data.frame which should content a column named "dir".
---	---

Details

Angle argument of Arrowhead function from "shape" package needs to be fed in an anti-clockwise way, relative to x-axis, in degrees 0,360. arrowDir function adapts wind direction provided by wind.fit (clockwise, relative to y-axis) to requirements of Arrowhead.

Value

A vector with angles for each arrow to be plotted by Arrowhead.

Note

arrowDir function works always together with Arrowhead function from "shape" package.

Author(s)

Javier Fernández-López

References

Karline Soetaert (2017). shape: Functions for Plotting Graphical Shapes, Colors. R package version 1.4.3. <https://CRAN.R-project.org/package=shape>

See Also[wind.dl](#)**Examples**

```

data(wind.data)

# Create a vector with wind direction (angles) adapted
alpha <- arrowDir(wind.data)
## Not run:
# Now, you can plot wind direction with Arrowhead function from shapes package
# Load "shape" package
require(shape)
plot(wind.data$lon, wind.data$lat, type = "n")
Arrowhead(wind.data$lon, wind.data$lat,
  angle = alpha,
  arr.length = 0.1, arr.type = "curved"
)

## End(Not run)

```

cost.active

*Relative cost of active movement through a flow***Description**

cost.active() is a deliberately simple, general index based on signed flow support. The flow component parallel to movement is $\text{speed} * \cos(\text{angle})$. Supportive flow lowers cost, opposing flow raises it, and perpendicular flow has no effect. This is a relative index, not physical travel time or energy.

Usage

```
cost.active(flow.direction, flow.speed, target, speed.scale = NULL)
```

Arguments

flow.direction	Flow direction in degrees, pointing towards where the flow moves.
flow.speed	Non-negative flow speed.
target	Movement direction in degrees.
speed.scale	Positive reference speed used to make flow support dimensionless. By default, the median positive input speed is used.

Details

Let $z = \text{speed} * \cos(\text{angle}) / \text{speed.scale}$. When $z \geq 0$, cost is $1 / (1 + z)$; when $z < 0$, cost is $1 - z$. Thus support equal to the reference speed halves cost, opposition equal to the reference speed doubles it, and $\text{cost}(-z) = 1 / \text{cost}(z)$.

Value

Relative movement cost. A value below one indicates support and a value above one indicates opposition.

cost.FMGS	<i>Compute flow-based cost or conductance</i>
-----------	---

Description

flow.dispersion computes movement conductance through a flow such as sea or wind currents. Passive movement uses a restricted version of the formula described in Felícisimo et al. (2008):

Usage

```
cost.FMGS(
  wind.direction,
  wind.speed,
  target,
  type = c("passive", "active"),
  speed.scale = NULL
)

flow.dispersion(
  x,
  fun = cost.FMGS,
  output = "transitionLayer",
  type = c("passive", "active"),
  speed.scale = NULL,
  ...
)
```

Arguments

wind.direction	A vector or scalar containing wind directions.
wind.speed	A vector or scalar containing wind speeds.
target	direction of the target cell
type	Movement model. In "passive" mode, deviations of 90 degrees or more from the flow direction are forbidden. In "active" mode, supportive flow lowers relative cost and opposing flow raises it; movement against the flow remains possible.
speed.scale	Positive reference speed for the active model. If NULL, the median positive speed of the complete raster layer is used. For comparisons among dates or regions, supply one fixed value in the same units as wind.speed.
x	A SpatRaster returned by wind2raster() with direction and speed layers, a legacy RasterStack, or a list of either type.

fun	A function to compute the cost to move between cells. The default is <code>cost.FMGS</code> from Felicísimo et al. (2008), see details.
output	This argument allows to select different kinds of output. "raw" mode creates a matrix (class "dgCMatrix") with transition costs between all cells in the raster. "transitionLayer" creates a TransitionLayer object with conductance values to be used with "gdistance" package.
...	Further arguments passed to or from other methods.

Details

$$\text{Cost} = (1/\text{Speed}) * (\text{HorizontalFactor})$$

where HorizontalFactor incrementally penalizes angular deviations from the flow direction and prevents passive movement at deviations of 90 degrees or more. Active movement uses a general relative index based on signed flow support rather than the original formula.

Value

In "transitionLayer" output, the function returns conductance values ($1/\text{cost}$) to move between all cells in a raster having into account flow speed and direction obtained from `wind.fit` function ("rWind" package). As wind or sea currents implies directionality, `flow.dispersion` produces an anisotropic conductance matrix (asymmetric). Conductance values are used later to built a TransitionLayer object from "gdistance" package.

In "raw" output, `flow.dispersion` creates a sparse Matrix with cost values. Each edge cost is the mean of the local costs at its origin and destination. Diagonal edges are multiplied by $\sqrt{2}$.

Note

The active model is a relative connectivity index, not a mechanistic model of animal movement. It ignores the component of flow perpendicular to movement.

Costs are relative per grid step. The diagonal correction assumes a square grid, but no correction is made for the changing physical distances of longitude/latitude cells. Do not interpret results as physical time, energy, or distance or compare them directly across grid resolutions or distant latitude ranges.

Note that for large data sets, it could take a while. For large study areas is strongly advised perform the analysis in a remote computer or a cluster.

Author(s)

Javier Fernández-López; Klaus Schliep; Yurena Arjona

References

- Felicísimo, Á. M., Muñoz, J., & González-Solis, J. (2008). Ocean surface winds drive dynamics of transoceanic aerial movements. *PLoS One*, 3(8), e2928.
- Jacob van Etten (2017). R Package gdistance: Distances and Routes on Geographical Grids. *Journal of Statistical Software*, 76(13), 1-21. doi:10.18637/jss.v076.i13

See Also[wind.dl](#), [wind2raster](#)**Examples**

```
require(gdistance)

data(wind.data)

wind <- wind2raster(wind.data)

Conductance <- flow.dispersion(wind, type = "passive")

transitionMatrix(Conductance)
image(transitionMatrix(Conductance))
```

flow.dispersion_int	<i>Compute flow-based cost or conductance</i>
---------------------	---

Description

flow.dispersion_int computes movement conductance through a flow such as sea or wind currents. Passive movement uses a restricted version of the formula described in Felícisimo et al. (2008):

Usage

```
flow.dispersion_int(
  stack,
  fun = cost.FMGS,
  output = "transitionLayer",
  type = c("passive", "active"),
  speed.scale = NULL,
  ...
)
```

Arguments

stack	A <code>SpatRaster</code> returned by <code>wind2raster()</code> with direction and speed layers. Legacy <code>RasterStack</code> objects are also accepted.
fun	A function to compute the cost to move between cells. The default is <code>cost.FMGS</code> from Felícisimo et al. (2008), see details.
output	This argument allows to select different kinds of output. "raw" mode creates a matrix (class " <code>dgCMatrix</code> ") with transition costs between all cells in the raster. "transitionLayer" creates a <code>TransitionLayer</code> object with conductance values to be used with "gdistance" package.
type	Movement model: "passive" or "active".

speed.scale	Positive reference speed for the active model. If NULL, the median positive speed of the complete raster layer is used.
...	Further arguments passed to or from other methods.

Details

$\text{Cost} = (1/\text{Speed}) * (\text{HorizontalFactor})$

where HorizontalFactor incrementally penalizes angular deviations from the flow direction and prevents passive movement at deviations of 90 degrees or more. Active movement instead uses the signed flow component parallel to the movement direction; supportive flow lowers cost and opposing flow raises it.

Value

In "transitionLayer" output, the function returns conductance values ($1/\text{cost}$) to move between all cells in a raster having into account flow speed and direction obtained from wind.fit function("rWind" package). As wind or sea currents implies directionality, flow.dispersion produces an anisotropic conductance matrix (asymmetric). Conductance values are used later to built a TransitionLayer object from "gdistance" package.

In "raw" output, flow.dispersion creates a sparse Matrix with cost values. Each edge cost is the mean of the local costs at its origin and destination. Diagonal edges are multiplied by $\sqrt{2}$.

Note

Costs are relative per grid step. The diagonal correction assumes a square grid, but no correction is made for the changing physical distances of longitude/latitude cells. Results should not be interpreted as physical time, energy, or distance and should not be compared directly across grid resolutions or distant latitude ranges.

Note that for large data sets, it could take a while. For large study areas is strongly advised perform the analysis in a remote computer or a cluster.

Author(s)

Javier Fernández-López; Klaus Schliep; Yurena Arjona

References

Felicísimo, Á. M., Muñoz, J., & González-Solis, J. (2008). Ocean surface winds drive dynamics of transoceanic aerial movements. PLoS One, 3(8), e2928.

Jacob van Etten (2017). R Package gdistance: Distances and Routes on Geographical Grids. Journal of Statistical Software, 76(13), 1-21. doi:10.18637/jss.v076.i13

See Also

[wind.dl](#), [wind2raster](#)

Examples

```
data(wind.data)
wind <- wind2raster(wind.data)
Conductance <- flow.dispersion(wind, type = "passive")
## Not run:
require(gdistance)
transitionMatrix(Conductance)
image(transitionMatrix(Conductance))

## End(Not run)
```

 sea0scar.dl

OSCAR Sea currents data download

Description

seaOscar.dl downloads sea currents data from the Ocean Surface Current Analyses Real-time (OSCAR) (PO.DAAC, [doi:10.5067/OSCAR03D01](https://doi.org/10.5067/OSCAR03D01)). Geospatial resolution is 0.33 degrees and sea currents are calculated for 15 m depth. Longitudes in the usual -180 to 180 notation are translated to the 20 to 420 degree axis used by the service.

Usage

```
sea0scar.dl(
  yyyy,
  mm,
  dd,
  lon1,
  lon2,
  lat1,
  lat2,
  type = "read-data",
  trace = 1
)
```

Arguments

yyyy	Selected year.
mm	Selected month.
dd	Selected day.
lon1	Western longitude
lon2	Eastern longitude
lat1	Northern latitude
lat2	Southern latitude

type	Output type. "read-data" is selected by default, creating an R object. If you choose "csv", seaOscar.dl create a a CSV file in your working directory named "oscar_yyyy_mm_dd.csv".
trace	if trace = 1 (by default) track downloaded files

Details

The output type is determined by type="csv" or type="read-data". If type="csv" is selected, the function creates a "sea_yyyy_mm_dd.csv" file that is downloaded at the work directory. If type="read-data" is selected, an R object (data.frame) is created.

Value

"rWind" and "data.frame" class object or .csv file with U and V vector components and sea current direction and speed for each coordinate in the study area defined by lon1/lon2 and lat1/lat2.

Author(s)

Javier Fernández-López (jflopez.bio@gmail.com)

References

[doi:10.5067/OSCAR03D01](https://doi.org/10.5067/OSCAR03D01)

See Also

[wind.dl_2](#), [wind2raster](#)

Examples

```
# Download sea currents for Galapagos Islands
## Not run:

seaOscar.dl(2015, 1, 1, -93, -88, 2, -3)

## End(Not run)
```

tidy

Transforming a rWind_series object into a data.frame

Description

The output of tidy is always a data.frame. It is therefore suited for further manipulation by packages like dplyr, reshape2, ggplot2 and ggvis.

Usage

```
tidy(x, ...)

## S3 method for class 'rWind_series'
tidy(x, ...)
```

Arguments

x An object to be converted into a tidy data.frame

... extra arguments

Examples

```
data(wind.series)
df <- tidy(wind.series)
head(df)
## Not run:
# use the tidyverse
library(dplyr)
mean_speed <- tidy(wind.series) %>%
  group_by(lat, lon) %>%
  summarise(speed = mean(speed))
wind_average2 <- wind.mean(wind.series)
all.equal(wind_average2$speed, mean_speed$speed)

## End(Not run)
```

uv2ds

*Transform U and V components in direction and speed and vice versa***Description**

Transform U and V components in direction and speed and vice versa

Usage

```
uv2ds(u, v)

ds2uv(d, s)
```

Arguments

u U component.

v U component.

d direction (degrees).

s speed (m/s).

Value

"uv2ds" returns a matrix with direction and speed values

"ds2uv" returns a matrix with U and V values

Note

Multiple U and V values can be processed. "dir" denotes where the wind/sea current is going (toward), not from where is coming.

Author(s)

Javier Fernández-López (jflopez.bio@gmail.com)

See Also

[wind.mean](#), [wind2raster](#)

Examples

```
(ds <- uv2ds(c(1, 1, 3, 1), c(1, 1.7, 3, 1)))
ds2uv(ds[, 1], ds[, 2])
```

wind.data

Wind data example

Description

This is an example of wind data obtained with wind.dl function for the Iberian Peninsula coordinates on 12/February/2015 at 00:00 (UTC)

Format

A list with one data.frame with 651 observations on the following 7 variables:

list("time (UTC)") a numeric with selected time of wind data

list("latitude (degrees_north)") a numeric with latitude values

list("longitude (degrees_east)") a numeric with longitude values

list("ugrd10m (m s-1)") a numeric with U component of wind data

list("vgrd10m (m s-1)") a numeric with V component of wind data

list("dir") a numeric with direction of wind data

list("speed") a numeric with speed of wind data

Details

This data set is the result of:

```
wind.data <- wind.dl(2015, 2, 12, 0, -10, 5, 35, 45)
```

Source

<https://allthiswasfield.blogspot.com/2016/12/rwind-r-package-released.html>

References

<https://www.ncei.noaa.gov/products/weather-climate-models/global-forecast>

Examples

```
data(wind.data)
str(wind.data)
head(wind.data[[1]])
```

wind.dl

Wind-data download

Description

wind.dl downloads wind data from the Global Forecast System (GFS) of the USA's National Weather Service (NWS) (<https://www.ncei.noaa.gov/products/weather-climate-models/global-forecast>). Wind data are taken from NOAA/NCEP Global Forecast System (GFS) Atmospheric Model collection. Geospatial resolution is 0.5 degrees (approximately 50 km), and wind is calculated for Earth surface, at 10 m. More metadata information for the current PacIOOS data: https://pae-paha.pacioos.hawaii.edu/erddap/griddap/ncep_global.graph Historical data are obtained from the official NOAA/NCEI GFS 0.5 degree archive: <https://www.ncei.noaa.gov/access/metadata/landing-page/bin/iso?id=gov.noaa.ncdc:C00634>

Usage

```
wind.dl(
  yyyy,
  mm,
  dd,
  tt,
  lon1,
  lon2,
  lat1,
  lat2,
  type = "read-data",
  trace = 1,
  source = c("auto", "pacioos", "ncei")
)

read.rWind(file)
```

Arguments

yyyy	Selected year.
mm	Selected month.
dd	Selected day.
tt	Selected time. There are currently several options at the GFS database: 00:00 - 03:00 - 06:00 - 09:00 - 12:00 - 15:00 - 18:00 - 21:00 (UTC).
lon1	Western longitude
lon2	Eastern longitude
lat1	Southern latitude
lat2	Northern latitude
type	Output type. "read-data" is selected by default, creating an R object. If you choose "csv", wind.dl create a a CSV file in your working directory named "wind_yyyy_mm_dd_tt.csv".
trace	if trace = 1 (by default) track downloaded files
source	Data provider. "auto" selects PacIOOS for dates in its current coverage and NOAA/NCEI for older dates. Use "pacioos" or "ncei" to force one provider.
file	file name of the saved ".csv" files.

Details

The output type is determined by type="csv" or type="read-data". If type="csv" is selected, the function creates a "wind_yyyy_mm_dd_tt.csv" file that is downloaded at the work directory. If type="read-data" is selected, an R object (data.frame) is created.

Value

"rWind" and "data.frame" class object or .csv file with U and V vector components and wind direction and speed for each coordinate in the study area defined by lon1/lon2 and lat1/lat2.

Note

Longitude coordinate are provided by GFS dataset in 0/360 notation and transformed internally into -180/180. Wind "dir" denotes where the wind is going (toward), not from where is coming. With source = "auto", dates before the current PacIOOS coverage are requested from NOAA/NCEI.

Author(s)

Javier Fernández-López (jflopez.bio@gmail.com)

References

<http://www.digital-geography.com/cloud-gis-getting-weather-data/#.WDOWmbV1DCL>
https://pae-paha.pacioos.hawaii.edu/erddap/griddap/ncep_global.graph

See Also[wind.dl_2](#), [wind2raster](#)**Examples**

```
# Download wind for Iberian Peninsula region at 2015, February 12, 00:00
## Not run:

wind.dl(2015, 2, 12, 0, -10, 5, 35, 45)

## End(Not run)
```

wind.dl_2

*Wind-data download***Description**

wind.dl_2 downloads time-series wind data from the Global Forecast System (GFS) of the USA's National Weather Service (NWS) (<https://www.ncei.noaa.gov/products/weather-climate-models/global-forecast>). Wind data are taken from NOAA/NCEP Global Forecast System (GFS) Atmospheric Model collection. Geospatial resolution is 0.5 degrees (approximately 50 km), and wind is calculated for Earth surface, at 10 m. More metadata information for current data: <https://pae-paha.pacioos.hawaii.edu/erddap/grid>. Historical data are obtained from the official NOAA/NCEI GFS 0.5 degree archive.

Usage

```
wind.dl_2(
  time,
  lon1,
  lon2,
  lat1,
  lat2,
  type = "read-data",
  trace = 1,
  source = c("auto", "pacioos", "ncei")
)

## S3 method for class 'rWind_series'
x[[i, exact = TRUE]]
```

Arguments

time	a scalar or vector of POSIXt or Date objects or an character which can transformed into those, see example below. There are currently these options at the GFS database for the hours: 00:00 - 03:00 - 06:00 - 09:00 - 12:00 - 15:00 - 18:00 - 21:00 (UTC) (TO).
------	--

lon1	Western longitude
lon2	Eastern longitude
lat1	Southern latitude
lat2	Northern latitude
type	Output type. "read-data" is selected by default, creating an R object. If you choose "csv", wind.dl create a a CSV file in your work directory named "wind_yyyy_mm_dd_tt.csv".
trace	if trace = 1 (by default) track downloaded files
source	Data provider. "auto" selects PacIOOS for dates in its current coverage and NOAA/NCEI for older dates. Use "pacioos" or "ncei" to force one provider.
x	object from which to extract element(s).
i	indices specifying elements to extract.
exact	Controls possible partial matching (not used yet).

Details

To get the same format as wind.dl, you should run tidy function from wind.dl_2 output. The output type is determined by type="csv" or type="read-data". If type="csv" is selected, the function creates a "wind_yyyy_mm_dd_tt.csv" file that is downloaded at the work directory. If type="read-data" is selected, an rWind_series object is created.

Value

an object of class rWind_series or .csv file/s with U and V vector components and wind direction and speed for each coordinate in the study area defined by lon1/lon2 and lat1/lat2.

Note

wind.dl_2 requires two dates that represent the boundaries of the time lapse to download wind series data. U and V vector components allow you to create wind averages or tendencies for each coordinate at the study area. Longitude coordinates are provided by GFS dataset in 0/360 notation and transformed internally into -180/180. "dir" denotes where the wind/sea current is going (toward), not from where is coming. With source = "auto", dates before the current PacIOOS coverage are requested from NOAA/NCEI.

Author(s)

Javier Fernández-López (jflopez.bio@gmail.com)

References

<http://www.digital-geography.com/cloud-gis-getting-weather-data/#.WDOWmbV1DCL>
https://pae-paha.pacioos.hawaii.edu/erddap/griddap/ncep_global.graph

See Also

[wind.mean](#), [wind2raster](#), [wind.dl](#), [as_datetime](#), [as.POSIXct](#)

Examples

```
# Download wind for Iberian Peninsula region at 2015, February 12, 00:00
## Not run:

wind.dl_2("2018/3/15 9:00:00", -10, 5, 35, 45)

library(lubridate)
dt <- seq(ymd_hms(paste(2018, 1, 1, 00, 00, 00, sep = "-")),
  ymd_hms(paste(2018, 1, 2, 21, 00, 00, sep = "-")),
  by = "3 hours"
)
ww <- wind.dl_2(dt, -10, 5, 35, 45)
tidy(ww)

## End(Not run)
```

wind.fit_int

wind.fit_int

Description

wind.fit_int is used internally by wind.dl to transform downloaded data from GFS. wind.fit_int applies trigonometry tools to transform U and V vector wind components in wind direction and speed features. It also transforms 0-360 longitude notation obtained from GFS data into -180/180 longitude notation. Moreover, it cleans dates names and sorts the data by latitude.

Usage

```
wind.fit_int(tmpx)
```

Arguments

tmpx downloaded data produced by wind.dl.

Value

data.frame

Note

This function is used internally by wind.dl

Author(s)

Javier Fernández-López (jflopez.bio@gmail.com)

References

https://en.wikipedia.org/wiki/Cross_product

See Also

[wind.dl](#), [wind.mean](#), [wind2raster](#)

wind.mean

Wind-data mean

Description

wind.mean computes the mean (average) wind speed and wind direction of a time series dataset of winds of the same region. Summaries of time series are not trivial to compute. We compute the arithmetic mean for the wind speed. The direction as the circular mean, see https://en.wikipedia.org/wiki/Circular_mean for more details. The U and V components are afterwards transformed from these values.

Usage

```
wind.mean(x)
```

Arguments

x An object of class rWind_series

Value

An object of class rWind, which is a data.frame

Note

For large time series, it could take a while.

Author(s)

Javier Fernández-López (jflopez.bio@gmail.com)

References

https://en.wikipedia.org/wiki/Cross_product

See Also

[wind.dl](#)

Examples

```
data(wind.series)
wind_average <- wind.mean(wind.series)
```

wind.series

*Wind series example***Description**

This is an example of a wind series data obtained with `wind.dl` function for New Zealand area on 3/January/2015 at all the available times: 00:00 - 03:00 - 06:00 - 09:00 - 12:00 - 15:00 - 18:00 - 21:00 (UTC)

Format

The format is an `rWind` list of 8 `data.frame`. Each `data.frame` contain 961 observations on the following 7 variables:

list("time (UTC)") a factor with selected time of wind data

list("latitude (degrees_north)") a factor with latitude values

list("longitude (degrees_east)") a factor with longitude values

list("ugrd10m (m s-1)") a factor with U component of wind data

list("vgrd10m (m s-1)") a factor with V component of wind data

list("dir") a numeric with direction of wind data

list("speed") a numeric with speed of wind data

Details

This data set is the result of:

```
library(lubridate) dt <- seq(ymd_h(paste(2015,1,3,00, sep="-")), ymd_h(paste(2015,1,3,21,
sep="-")), by="3 hours")
wind.series <- wind.dl_2(dt, 164, 179, -48, -33)
```

Source

<https://allthiswasfield.blogspot.com/2016/12/rwind-r-package-released.html>

References

<https://www.ncei.noaa.gov/products/weather-climate-models/global-forecast>

Examples

```
data(wind.series)
str(tidy(wind.series))
```

wind2raster	<i>Wind data to terra rasters</i>
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Description

`wind2raster()` converts an `rWind` object to a two-layer `terra::SpatRaster` containing wind direction and speed. An `rWind_series` is converted to a list of `SpatRaster` objects, one per time step. All outputs use WGS 84 (EPSG:4326).

Usage

```
wind2raster(x)
```

Arguments

`x` An object of class `rWind` or `rWind_series`.

Value

A `terra::SpatRaster`, or a list of them, with layers direction and speed.

Author(s)

Javier Fernández-López (jflopez.bio@gmail.com)

See Also

[wind.dl](#)

Examples

```
data(wind.data)

# Create a SpatRaster with wind direction and speed layers

wind2raster(wind.data)
```

wind2raster_int	<i>Wind data to a terra raster</i>
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Description

wind2raster_int() converts one gridded rWind data frame to a two-layer [terra::SpatRaster](#). Longitude and latitude define the grid and the output layers contain wind direction and speed. The coordinate reference system is WGS 84 (EPSG:4326).

Usage

```
wind2raster_int(x)
```

Arguments

x	an object of class rWind
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Value

A two-layer [terra::SpatRaster](#) named direction and speed.

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See Also

[wind.dl](#), [wind2raster](#)

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